Supplementary Material

One-shot screening of linear preprocessing operators for PLS and Ridge calibration in near-infrared spectroscopy

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Contents

1 Notation, Scope and Complete Derivations

1.1 Strict linear operators

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a spectral matrix and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ a response matrix. A strict linear spectral operator $\mathbf{A}_b \in \mathbb{R}^{p \times p}$ acts on row spectra as

$$\mathbf{X}_b = \mathbf{X}\mathbf{A}_b^{\top}. \tag{1}$$

The operator is fixed once the wavelength grid and hyperparameters are fixed. The compact bank contains identity, Savitzky–Golay smoothers and derivatives, finite differences and polynomial detrending projections.

Sample-adaptive or fitted transformations do not satisfy this definition. SNV uses sample-specific statistics, MSC and EMSC estimate correction parameters relative to a reference, and ASLS solves an asymmetric baseline problem. They are valid preprocessing branches, but they are fitted inside each calibration fold and are not inserted into the strict-linear operator identities.

1.2 AOM-PLS covariance identity

For centered $\mathbf{X}$ and $\mathbf{Y}$, standard PLS can be written with weights $\mathbf{W}$, loadings $\mathbf{P}$ and response loadings $\mathbf{Q}$:

$$\mathbf{B} = \mathbf{W}(\mathbf{P}^{\top}\mathbf{W})^{+}\mathbf{Q}^{\top}. \tag{2}$$

For a fixed operator $\mathbf{A}_b$,

$$(\mathbf{X}\mathbf{A}_b^{\top})^{\top}\mathbf{Y} = \mathbf{A}_b\mathbf{X}^{\top}\mathbf{Y}. \tag{3}$$

Equation 3 is the algebraic reason why strict-linear preprocessing alternatives can be evaluated in covariance space. The SIMPLS-covariance implementation acts on $\mathbf{X}^{\top}\mathbf{Y}$ directly. The NIPALS-adjoint implementation applies $\mathbf{A}_b$ and $\mathbf{A}_b^{\top}$ during the iterative extraction without storing every transformed design matrix.

If a transformed-space direction for component a is r_a, the original-wavelength direction is

$$z_a = \mathbf{A}_b^{\top} r_a. \tag{4}$$

With $\mathbf{Z} = [z_1, \ldots, z_k]$ and $\mathbf{T} = \mathbf{X}\mathbf{Z}$, the final coefficient matrix is

$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^{\top}\mathbf{Z})^{+}\mathbf{Q}^{\top}. \tag{5}$$

Thus the fitted AOM-PLS object predicts from original spectra and can expose coefficients on the original wavelength grid.

1.3 AOM-Ridge kernels

For centered $\mathbf{X}_c$ and $\mathbf{Y}_c$, a single operator induces

$$\mathbf{Z}_b = \mathbf{X}_c\mathbf{A}_b^{\top}, \tag{6}$$

$$\mathbf{K}_b = \mathbf{Z}_b\mathbf{Z}_b^{\top} = \mathbf{X}_c\mathbf{A}_b^{\top}\mathbf{A}_b\mathbf{X}_c^{\top}, \tag{7}$$

$$\mathbf{C}_b = (\mathbf{K}_b + \alpha\mathbf{I}_n)^{-1}\mathbf{Y}_c, \tag{8}$$

and the recovered original-space coefficient matrix is

$$\boldsymbol{\beta}_b = \mathbf{A}_b^{\top}\mathbf{A}_b\mathbf{X}_c^{\top}\mathbf{C}_b. \tag{9}$$

For weighted operator mixtures,

$$\mathbf{K} = \sum_{b=1}^{m} s_b^2\mathbf{X}_c\mathbf{A}_b^{\top}\mathbf{A}_b\mathbf{X}_c^{\top}, \qquad \mathbf{M} = \sum_{b=1}^{m} s_b^2\mathbf{A}_b^{\top}\mathbf{A}_b, \tag{10}$$

so

$$\boldsymbol{\beta} = \mathbf{M}\mathbf{X}_c^\top(\mathbf{K} + \alpha\mathbf{I}_n)^{-1}\mathbf{Y}_c. \tag{11}$$

This is the Ridge counterpart of AOM-PLS: PLS adapts supervised cross-covariance directions, while Ridge adapts the regularized sample geometry.

1.4 Ridge selector family

The main manuscript now reports both the plain global compact selector and the Blender selector. The full Ridge family also includes branch-specific compact variants, local-neighbourhood selectors and multi-branch kernel mixtures. Global selectors choose one operator geometry for the whole calibration set. Local selectors restrict the geometry to neighbourhoods such as $k = 50$ nearest calibration samples. Multi-branch kernels combine several branch geometries with shrinkage. These variants are useful for understanding the design space, but their refreshed paired statistics do not beat Blender on the main Ridge denominators.

1.5 FastAOM derivation

FastAOM is treated here as an exploration of alternative AOM forms, not as a main result. It defines a chain as an ordered composition of primitive strict operators. The grammar assigns roles to operators: baseline/projection, smoothing, derivative and projection-mask operations. A depth cap limits the number of composed operators, and grammar pruning removes invalid or redundant role sequences before screening.

For each fold-local base $B_j(\mathbf{X})$ and chain s, FastAOM computes

$$\text{score}(j, s) = \frac{\left\| A_s B_j(\mathbf{X}_c)^\top y_c \right\|_2^2}{\|B_j(\mathbf{X}_c)A_s^\top\|_F^2 \, \|y_c\|_2^2}. \tag{12}$$

With a truncated SVD $B_j(\mathbf{X}_c) \approx U_j\Sigma_j V_j^\top$ and $F_{j,s} = A_s V_j \Sigma_j$, the screened kernel is approximated by

$$\mathbf{K}_{j,s} = B_j(\mathbf{X}_c)A_s^\top A_s B_j(\mathbf{X}_c)^\top \approx U_j C_{j,s} U_j^\top, \qquad C_{j,s} = F_{j,s}^\top F_{j,s}. \tag{13}$$

Table 1: FastAOM model families. These are supplementary explorations of alternative AOM forms.

Class	Routing strategy	Role in this work
`SingleChainPLSRidge`	Top screened chain followed by PLS-Ridge	Fast baseline equivalent to selecting one screened chain.
`HardAOMChainPLSRidge`	One greedy chain per latent component	Diagnostic component-level routing.
`SoftAOMChainPLSRidge`	Sparse non-negative chain mixture per component	Mixture-routing exploration.
`SparseMultiKernelRidge`	Greedy sparse non-negative kernel mixture	Best FastAOM exploration family in the wide seed0 table.
`FastAOMPLSRidge`	Orchestrator over bases, chains, SVD, screening and fit	Entry point used by the benchmark runner.

2 Cohort Manifest

The refreshed manifest contains 61 regression rows and 17 classification manifest rows. One classification row is retained only as an excluded manifest record because the source files are missing, leaving 16 classification rows included in benchmark analyses. The overview longtable below lists the full manifest for provenance. The per-dataset regression score table in Section 8 is restricted to the strict main intersection $N_\cap = 32$. The classification tables report their own paired denominators.

Table 2: Cohort denominator summary.

Property	Summary used in this work
Regression manifest	61 included regression rows across the benchmark inventory.
Classification manifest	17 classification manifest rows; 16 included rows after one missing-file exclusion.
Main regression denominator	$N_\cap = 32$ datasets for the eight paper variants.
Paper variants	PLS-default, PLS-HPO, AOM-PLS (simple), AOM-PLS (best), Ridge-default, Ridge-HPO, AOM-Ridge (simple) and AOM-Ridge (best).
Analytical domains	Leaf physiology, fruit quality, grain and seed traits, dairy, beverages, meat quality, petroleum, soil, manure, wood products, tablets and public calibration datasets.
Sample and wavelength range	Training sets span 28–39,225 samples and 125–4,200 spectral variables in the local regression manifest.
Validation rule	External test sets are never used for preprocessing, operator, component or regularization selection.

Table 3: Full manifest overview for the AOM benchmark cohort. This table is for provenance and includes manifest rows beyond the main $N_\cap = 32$ score denominator; the per-dataset score table is restricted to the paired denominator stated in the text. The longtable repeats the header after page breaks and marks rows continued on the next page.

Dataset	Task	n	p	p/n	Response type or range	Original split	Domain
CoffeeType_kenstone70_strat	classification	70	601	8.586	7 classes; max share 0.14	kenstone70_ strat	beverage
Species_56_Bagnall	classification	56	286	5.107	2 classes; max share 0.52	Bagnall	beverage
Oocist2C_333_Maia_Acc87.6	classification	333	2151	6.459	2 classes; max share 0.52	Maia	biomedical
Sporozoite2C_229_Maia_Acc94.5	classification	229	2151	9.393	2 classes; max share 0.60	Maia	biomedical
CT2C_1057_CIAT_Acc	classification	1056	1050	0.994	2 classes; max share 0.73	CIAT	crop-tuber
labels_kenstone70_strat	classification	450	601	1.336	9 classes; max share 0.11	kenstone70_ strat	dairy
Strawberry2C_983_Holland_Acc94.3	classification	983	235	0.239	2 classes; max share 0.64	Holland	fruit-quality
Beef_Impurity_60_AlJowder	classification	60	470	7.833	5 classes; max share 0.20	AlJowder	meat-quality
FinalScoreBin_grp70_30_classStrat	classification	935	2177	2.328	2 classes; max share 0.58	grp70_30_ classStrat	plant-disease
ScoreBin_grp70_30_classStrat	classification	816	2177	2.668	2 classes; max share 0.77	grp70_30_ classStrat	plant-disease
Genotype10_250	classification	250	2152	8.608	10 classes; max share 0.10	unspecified	plant-id
Group9_1856	classification	1800	2152	1.196	9 classes; max share 0.16	unspecified	plant-id
Group_2185	classification	2185	2152	0.985	10 classes; max share 0.18	unspecified	plant-id
InOut_1264	classification	1263	2152	1.704	2 classes; max share 0.59	unspecified	plant-id
Species_code_grpStrat70_30_ bySpecimen	classification	7323	1951	0.266	5 classes; max share 0.31	grpStrat70_30	plant-id
C2_511_Davrieux_Acc82	classification	511	1050	2.055	2 classes; max share 0.51	Davrieux	wood-product
C5_511_Davrieux_Acc82	classification	511	1050	2.055	5 classes; max share 0.42	Davrieux	wood-product
Beer_OriginalExtract_60_KS	regression	60	576	9.600	Beer_OriginalExtract_60; range 4.23 to 18.8	KS	beverage
Beer_OriginalExtract_60_YbaseSplit	regression	60	576	9.600	Beer_OriginalExtract_60; range 4.23 to 18.8	YbaseSplit	beverage

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Dataset	Task	n	p	p/n	Response type or range	Original split	Domain
Malaria_Oocist_333_Maia	regression	333	2151	6.459	Malaria_Oocist_333; range 0 to 6.1e+04	Maia	biomedical
Malaria_Sporozoite_229_Maia	regression	229	2151	9.393	Malaria_Sporozoite_229; range 0 to 2.36e+05	Maia	biomedical
Corn_Oil_80_ ZhengChenPelegYbaseSplit	regression	80	700	8.750	Corn_Oil_80; range 3.09 to 3.83	YbaseSplit	crop-grain
Corn_Starch_80_ ZhengChenPelegYbaseSplit	regression	80	700	8.750	Corn_Starch_80; range 62.8 to 66.5	YbaseSplit	crop-grain
Rice_Amylose_313_YbasedSplit	regression	313	1154	3.687	Rice_Amylose_313; range 0 to 33.8	YbasedSplit	crop-grain
C_woOutlier	regression	2419	1154	0.477	C_woOutlier; range 21.5 to 47.5	woOutlier	crop-seed
N_wOutlier	regression	2427	1154	0.475	N_wOutlier; range 0.19 to 5.95	wOutlier	crop-seed
N_woOutlier	regression	2412	1154	0.478	N_woOutlier; range 0.19 to 5.95	woOutlier	crop-seed
Milk_Fat_1224_KS	regression	402	255	0.634	Milk_Fat_1224; range 1.54 to 7.6	KS	dairy
Milk_Lactose_1224_KS	regression	1224	255	0.208	Milk_Lactose_1224; range 3.98 to 5.1	KS	dairy
Milk_Urea_1224_KS	regression	1224	255	0.208	Milk_Urea_1224; range 9 to 44	KS	dairy
Biscuit_Fat_40_RandomSplit	regression	72	700	9.722	Biscuit_Fat_40; range 14.8 to 21.7	RandomSplit	food-product
Biscuit_Sucrose_40_RandomSplit	regression	72	700	9.722	Biscuit_Sucrose_40; range 9.95 to 23.2	RandomSplit	food-product
Brix_spxy70	regression	50	600	12.000	Brix; range 11.2 to 20	spxy70	fruit-quality
Firmness_spxy70	regression	40	600	15.000	Firmness; range 2.85 to 4.85	spxy70	fruit-quality
brix_groupSampleID_stratDateVar_ balRows	regression	2133	2101	0.985	brix; range 4.3 to 29	groupSampleID_ stratDateVar	fruit-quality
ph_groupSampleID_stratDateVar_ balRows	regression	1401	2101	1.500	ph; range 2.29 to 4.58	groupSampleID_ stratDateVar	fruit-quality
ta_groupSampleID_stratDateVar_ balRows	regression	1401	2101	1.500	ta; range 2.68 to 16.3	groupSampleID_ stratDateVar	fruit-quality
TIC_spxy70	regression	62	254	4.097	TIC; range 60.2 to 97.6	spxy70	industrial
ALPINE_P_291_KS	regression	291	2151	7.392	ALPINE_P_291; range -0.00724 to 0.696	KS	leaf-physiology
An_spxyG70_30_byCultivar_ASD	regression	112	2101	18.759	An; range 0.0565 to 16.5	spxyG_ byCultivar	leaf-physiology
An_spxyG70_30_byCultivar_ MicroNIR	regression	116	125	1.078	An; range 0.0565 to 18.6	spxyG_ byCultivar	leaf-physiology

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Dataset	Task	n	p	p/n	Response type or range	Original split	Domain
An__spxyG70__30__byCultivar__MicroNIR__NeoSpectra	regression	115	276	2.400	An; range 0.0565 to 18.6	spxyG__byCultivar	leaf-physiology
An__spxyG70__30__byCultivar__NeoSpectra	regression	119	257	2.160	An; range 0.0565 to 18.6	spxyG__byCultivar	leaf-physiology
Ccar__spxyG__block2deg	regression	4245	196	0.046	Ccar; range -999 to 28.4	spxyG__block2deg	leaf-physiology
Chla+b__spxyG__block2deg	regression	6850	196	0.029	Chla+b; range -999 to 167	spxyG__block2deg	leaf-physiology
Chla+b__spxyG__species	regression	6850	196	0.029	Chla+b; range -999 to 167	spxyG	leaf-physiology
LMA__spxyG70__30__byCultivar__ASD	regression	1564	2101	1.343	LMA; range 1.43 to 7.59	spxyG__byCultivar	leaf-physiology
LMA__spxyG__block2deg	regression	45417	196	0.004	LMA; range 0.066 to 389	spxyG__block2deg	leaf-physiology
LP__spxyG	regression	257	2101	8.175	LP; range 0.0681 to 0.671	spxyG	leaf-physiology
MP__spxyG	regression	257	2101	8.175	MP; range 0.0202 to 0.114	spxyG	leaf-physiology
NP__spxyG	regression	257	2101	8.175	NP; range 0.0561 to 0.57	spxyG	leaf-physiology
Pi__spxyG	regression	257	2101	8.175	Pi; range 0.0225 to 0.63	spxyG	leaf-physiology
Rd25__CBtestSite	regression	470	2151	4.577	Rd25__CBtestSite; range 0.465 to 2.03	external__site__CB	leaf-physiology
Rd25__GTtestSite	regression	470	2151	4.577	Rd25__GTtestSite; range 0.465 to 2.03	external__site__GT	leaf-physiology
Rd25__XSBNtestSite	regression	470	2151	4.577	Rd25__XSBNtestSite; range 0.465 to 2.03	external__site__XSBN	leaf-physiology
Rd25__spxy70	regression	470	2151	4.577	Rd25; range 0.465 to 2.03	spxy70	leaf-physiology
V25__spxyG	regression	250	2101	8.404	V25; range 0.0229 to 1.39	spxyG	leaf-physiology
WUEinst__spxyG70__30__byCultivar__MicroNIR__NeoSpectra	regression	112	276	2.464	WUEinst; range 0.833 to 12.4	spxyG__byCultivar	leaf-physiology
grapevine__chloride__556__KS	regression	555	1023	1.843	grapevine__chloride__556; range 0 to 7.95e+03	KS	leaf-physiology
Beef__Marbling__RandomSplit	regression	832	331	0.398	Beef__Marbling; range 100 to 810	RandomSplit	meat-quality
Quartz__spxy70	regression	303	1500	4.950	Quartz	spxy70	mineral
DIESEL__bp50__246__b-a	regression	226	401	1.774	DIESEL__bp50__246__b-a; range 197 to 293	unspecified	petroleum

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Dataset	Task	n	p	p/n	Response type or range	Original split	Domain
DIESEL_bp50_246_hla-b	regression	246	401	1.630	DIESEL_bp50_246_hla-b; range 197 to 293	unspecified	petroleum
DIESEL_bp50_246_hlb-a	regression	246	401	1.630	DIESEL_bp50_246_hlb-a; range 197 to 293	unspecified	petroleum
Escitalopramt_310_Zhao	regression	310	404	1.303	Escitalopramt_310; range 4.61 to 9.79	Zhao	pharmaceutical
FinalScore_grp70_30_scoreQ	regression	935	2177	2.328	FinalScore; range 1 to 5	grp70_30_scoreQ	plant-disease
Fv_Fm_grp70_30	regression	518	2177	4.203	Fv_Fm; range 0.512 to 0.856	grp70_30	plant-disease
Tleaf_grp70_30	regression	665	2177	3.274	Tleaf; range 19.7 to 44.7	grp70_30	plant-disease
All_manure_CaO_SPXY_strat_Manure_type	regression	490	1003	2.047	All_manure_CaO; range 1.53 to 146	SPXY_strat	soil-amendment
All_manure_K2O_SPXY_strat_Manure_type	regression	490	1003	2.047	All_manure_K2O; range 0.66 to 47.7	SPXY_strat	soil-amendment
All_manure_MgO_SPXY_strat_Manure_type	regression	490	1003	2.047	All_manure_MgO; range 0.52 to 18.8	SPXY_strat	soil-amendment
All_manure_P2O5_SPXY_strat_Manure_type	regression	490	1003	2.047	All_manure_P2O5; range 0.84 to 43.7	SPXY_strat	soil-amendment
All_manure_Total_N_SPXY_strat_Manure_type	regression	490	1003	2.047	All_manure_Total_N; range 2.07 to 40.5	SPXY_strat	soil-amendment
LUCAS_SOC_Cropland_8731_NocitaKS	regression	8731	4200	0.481	LUCAS_SOC_Cropland_8731; range 0 to 194	NocitaKS	soil-eu
LUCAS_SOC_all_26650_NocitaKS	regression	19036	4200	0.221	LUCAS_SOC_all_26650; range 0 to 587	NocitaKS	soil-eu
LUCAS_pH_Organic_1763_LiuRandomOrganic	regression	1763	4200	2.382	LUCAS_pH_Organic_1763; range 2.57 to 7.28	LiuRandom	soil-eu
WOOD_Density_402_Olale	regression	402	1038	2.582	WOOD_Density_402; range 0.197 to 0.952	Olale	wood-product
WOOD_N_402_Olale	regression	402	1038	2.582	WOOD_N_402; range 0.08 to 0.49	Olale	wood-product

2.1 Missingness audit

The strict paired denominator $N_\cap = 32$ used in the main text is the intersection of dataset/seed rows available for every paper variant. Table 4 reports, for each variant, how many of the 60–61 attempted regression rows produced complete results, how many failed with a fit error and how many were not attempted in the current workspace. The two recurring error datasets, `FinalScore_grp70_30_scoreQ` and `Tleaf_grp70_30`, contain NaN entries in the source spectra and fail under every linear regressor; they are therefore not specific to any variant. The remaining absences are dominated by the HPO rows, where the cartesian preprocessing grid was not run on the full cohort because of compute cost rather than dataset incompatibility. This is the conservative direction: the comparators that AOM is benchmarked against are the variants with the largest unfilled cohort, so the strict intersection narrows the comparison toward datasets where HPO had its best chance to win.

Table 4: Per-variant cohort coverage and absence reasons. "Complete" counts datasets with finite RMSEP across the requested seed budget; "Error" counts datasets where the fit raised an exception (almost always NaN in the source spectra); "Not attempted" counts datasets the workspace did not schedule.

Variant (paper label / runner key)	Complete	Error	Not attempted	Total available	Dominant reason for absence
PLS-default (`pls-default-cv5`)	57	3	1	60	NaN in source data
Ridge-default (`ridge-default-cv5`)	58	2	1	60	NaN in source data
PLS-HPO (`pls-tabpfn-hpo-25trials`)	36	2	23	38	Compute budget (not attempted)
Ridge-HPO (`ridge-tabpfn-hpo-60trials`)	35	2	24	37	Compute budget (not attempted)
AOM-PLS (simple, `AOM-compact-cv5`)	55	0	6	55	Workspace scope (not attempted)
AOM-PLS (best, `ASLS-AOM-compact-cv5`)	53	2	6	55	ASLS divergence / not attempted
AOM-Ridge (simple, `AOMRidge-global-compact-none`)	53	0	8	53	Workspace scope (not attempted)
AOM-Ridge (best, `AOMRidge-Blender-headline-spxy3`)	53	0	8	53	Workspace scope (not attempted)

3 Operator Bank Exploration

The compact bank contains nine operators:

$$\mathcal{A}_{\text{compact}} = \{\text{identity}, \mathrm{SG}^{(0)}_{11,2}, \mathrm{SG}^{(0)}_{21,3}, \mathrm{SG}^{(1)}_{11,2}, \mathrm{SG}^{(1)}_{21,3}, \\ \mathrm{SG}^{(2)}_{11,2}, \text{detrend}_1, \text{detrend}_2, \mathrm{FD}_1\}. \tag{14}$$

With B operators and $K_{\max}$ component prefixes, the selector considers approximately $BK_{\max}$ candidates before fold replication. At $K_{\max} = 15$, the compact bank has 135 operator/component candidates, whereas a 100-operator bank has 1500.

Table 5: Compact-bank operator frequencies from the refreshed selector diagnostics.

Compact-bank operator	Selections	Component fraction	Datasets using it
sg_smooth_w21_p3	876	23.4%	22
sg_d1_w21_p3	679	18.1%	24
sg_d1_w11_p2	497	13.3%	16
fd_d1	415	11.1%	14
detrend_d2	386	10.3%	16
identity	327	8.7%	11
detrend_d1	306	8.2%	18
sg_d2_w11_p2	151	4.0%	9
sg_smooth_w11_p2	112	3.0%	7

The wider Savitzky–Golay smoother `sg_smooth_w21_p3` is the most frequent selected operator (23.4% of selected components). First-derivative operators occupy the next two positions. Identity remains selected on 8.7% of selected components, confirming that the bank can fall back to raw centered spectra when preprocessing is not beneficial.

4 Full AOM-PLS Family

The main manuscript reports the plain compact-CV5 AOM-PLS baseline. It promotes the ASLS compact-CV5 branch as the selected PLS regression variant. Table 6 keeps the full AOM-PLS and POP-PLS family in the supplement. Ratios are computed against PLS-standard on matching dataset/seed rows.

Table 6: AOM-PLS family summary from the seeds012 run.

Variant	Datasets	Runs	Median RMSEP	Median ratio	Wins vs PLS
ASLS-AOM-compact-cv5-numpy	53	159	0.466	0.962	107/159
AOM-compact-cv3-numpy	55	165	0.442	0.980	98/165
AOM-compact-cv5-numpy	55	165	0.442	0.981	107/165
AOM-compact-simpls-covariance-numpy	55	165	0.795	0.999	84/165
AOM-default-nipals-adjoint-numpy	55	165	0.549	0.999	81/165
aom_nirs-AOM-PLS-default	55	165	0.549	0.999	81/165
PLS-standard-numpy	55	165	0.795	1.000	0/165
POP-nipals-adjoint-numpy	55	165	1.784	1.373	34/165
POP-simpls-covariance-numpy	55	165	1.805	1.385	33/165

The strict compact CV variants are close to the ASLS branch, but the ASLS branch has the best aggregate RMSEP ratio among the reported AOM-PLS rows. The default NIPALS-adjoint row is neutral at the aggregate level. POP regression variants are substantially weaker and are discussed separately below.

5 Full AOM-Ridge Family

The Ridge family includes headline Blender and AutoSelect rows, plus local and multi-branch variants. Table 7 reports the variant-level summary. Absolute RMSEP medians are descriptive only; the main choice is based on paired ratios and wins.

Table 7: AOM-Ridge family summary.

Source	Variant	Datasets	Runs	Median RMSEP	Median fit (s)
headline	AOMRidge-global-compact-none	53	53	0.359	21.60
headline	AOMRidge-global-compact-none-asls	53	53	0.382	24.35
headline	AOMRidge-global-compact-none-snv	53	53	0.389	18.23
headline	AOM-PLS-compact-CV	53	53	0.414	1.94
headline	AOMRidgePLS-compact-colscale-cv-relative	53	53	0.414	33.41
headline	AOMRidge-Blender-headline-spxy3	53	55	0.471	960.1
headline	AOMRidge-global-compact-none-msc	53	53	0.483	25.90
headline	AOMRidge-AutoSelect-headline-spxy3	53	55	0.520	646.2
headline	Ridge-raw	53	53	0.537	1.01
headline	AOMRidgePLS-compact-Hmax-relative-emsc2	53	53	1.981	29.96
seeds012	AOMRidge-global-compact-none-split_aware	25	75	0.371	24.68
seeds012	AOMRidge-Local-compact-cv-blended	25	75	0.484	4.19
seeds012	AOMRidge-Local-compact-knn50	25	75	0.523	3.19
seeds012	Ridge-raw	25	76	0.571	0.49
seeds012	AOMRidge-MultiBranchMKL-compact-shrink03	25	75	3.599	101.2

The global compact row is the plain Ridge AOM baseline used in the main text. Blender is promoted as the selected Ridge variant because it gives the best paired Ridge result against default Ridge and Ridge-HPO. AutoSelect has a similar median ratio against Ridge-HPO but fewer wins. Local selectors are fast and useful diagnostically, but they do not beat Ridge-HPO in the paired summary.

6 FastAOM Exploration

FastAOM remains a supplementary exploration. It tests whether low-rank screening and sparse kernel mixtures can explore a larger operator-chain space without fitting every chain as a full external pipeline.

The best FastAOM entry by median relative RMSEP after the $N \geq 50$ filter is `FastAOM-sparse-mkr-supervised`: $N = 50$, median relative RMSEP 1.009, 10 headline wins and median fit time 87.77 s. The compact sparse-MKR variant has median relative RMSEP 1.022 with median fit time 2.48 s. The compact single-chain variant has median relative RMSEP 1.052 and median fit time 1.86 s. These rows show that screened operator-chain models are promising, but the refreshed paired comparison does not beat the selected ASLS-AOM variant on the same wide table.

Table 8: FastAOM wide seed0 summary. Each row reports its own available complete datasets and is not the main $N_\cap = 32$ paired denominator.

Variant	Family	N	Median rel. RMSEP	Median fit (s)	Wins
FastAOM-sparse-mkr-supervised	Sparse MKR	50	1.009	87.77	10
ASLS-AOM-compact-cv5-numpy	Reference	57	1.011	1.36	8
AOM-compact-cv5-numpy	Reference	59	1.021	1.31	2
FastAOM-sparse-mkr-compact	Sparse MKR	50	1.022	2.48	3
nirs4all-AOM-PLS-default	Reference	59	1.034	0.97	7
FastAOM-single-chain-compact	Single chain	52	1.052	1.86	2
FastAOM-single-chain-compact-cv5-numpy	Single chain	52	1.052	2.02	0
PLS-standard-numpy	Reference	59	1.053	0.03	2
FastAOM-soft-chain-compact	Soft chain	52	1.062	3.05	1
FastAOM-hard-chain-osc	Hard chain	52	1.078	2.68	0
FastAOM-hard-chain-supervised	Hard chain	52	1.084	121.87	4
FastAOM-hard-chain-asls	Hard chain	52	1.105	174.46	3
FastAOM-hard-chain-multibase	Hard chain	52	1.109	5.62	0
FastAOM-hard-chain-compact	Hard chain	52	1.128	2.88	3
FastAOM-single-chain-supervised-cv5-numpy	Single chain	52	1.208	119.19	2
FastAOM-hard-chain-compact-d4	Hard chain	1	1.256	38.08	0

Hard-chain routing is valuable diagnostically because it assigns one chain per latent component. In the full table, however, the compact hard-chain variant has median relative RMSEP 1.128, and the ASLS, multibase and supervised hard-chain variants range from 1.084 to 1.109 with substantially higher runtime for the supervised and ASLS branches. These results explain why FastAOM is not promoted in the main manuscript.

7 POP-PLS Exploration

POP-PLS allows different operators across components, which is attractive as a modelling idea but weak in the current regression benchmark. The two regression POP variants have median ratios above 1.37 against PLS-standard and win on roughly one fifth of dataset/seed rows.

Table 9: POP-PLS regression summary from the AOM-PLS seeds012 run.

POP variant	Datasets	Runs	Median ratio vs PLS	Wins vs PLS
POP-nipals-adjoint-numpy	55	165	1.373	34/165
POP-simpls-covariance-numpy	55	165	1.385	33/165

The poor POP regression result is consistent with a noisy component-wise selection surface: extra flexibility increases variance without producing a stable aggregate gain. POP-style classification rows are kept in the full classification table, but POP is not part of the main result.

8 Per-Dataset Regression Results

The heatmap gives a compact per-dataset view, and the long table gives the underlying RMSEP values on the same $N_\cap = 32$ regression rows used in the main manuscript.

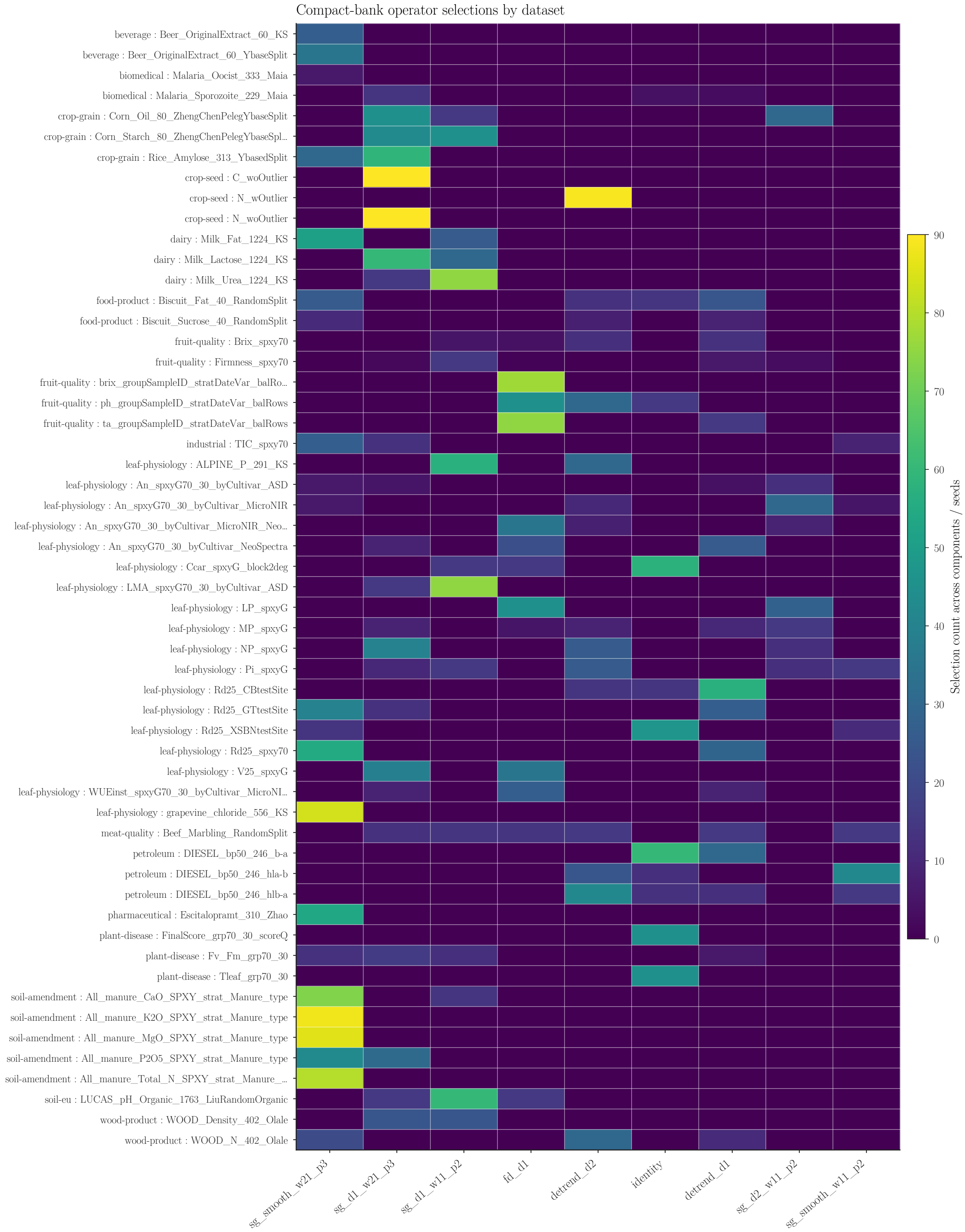

Figure 1: Per-dataset compact-bank operator heatmap. Cells count selected components across compact AOM-PLS variants and seeds.

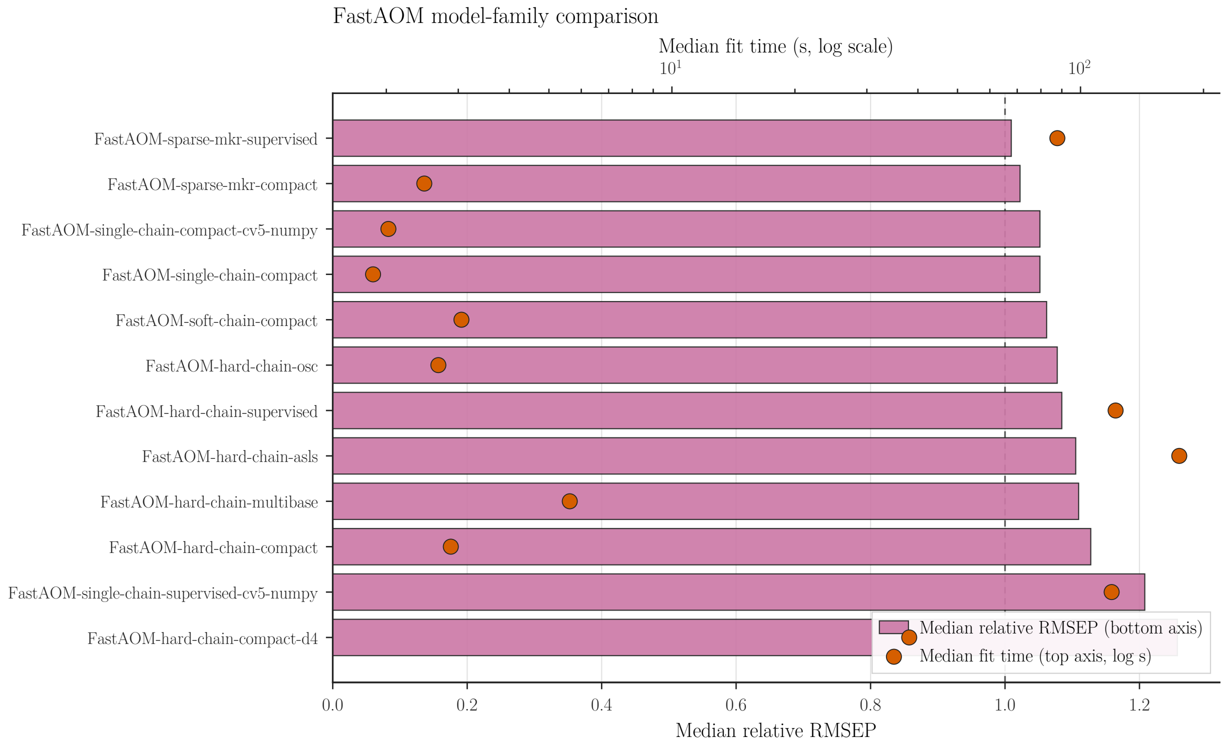

Figure 2: FastAOM variant comparison ordered by median relative RMSEP. The upper axis shows median fit time on a logarithmic scale.

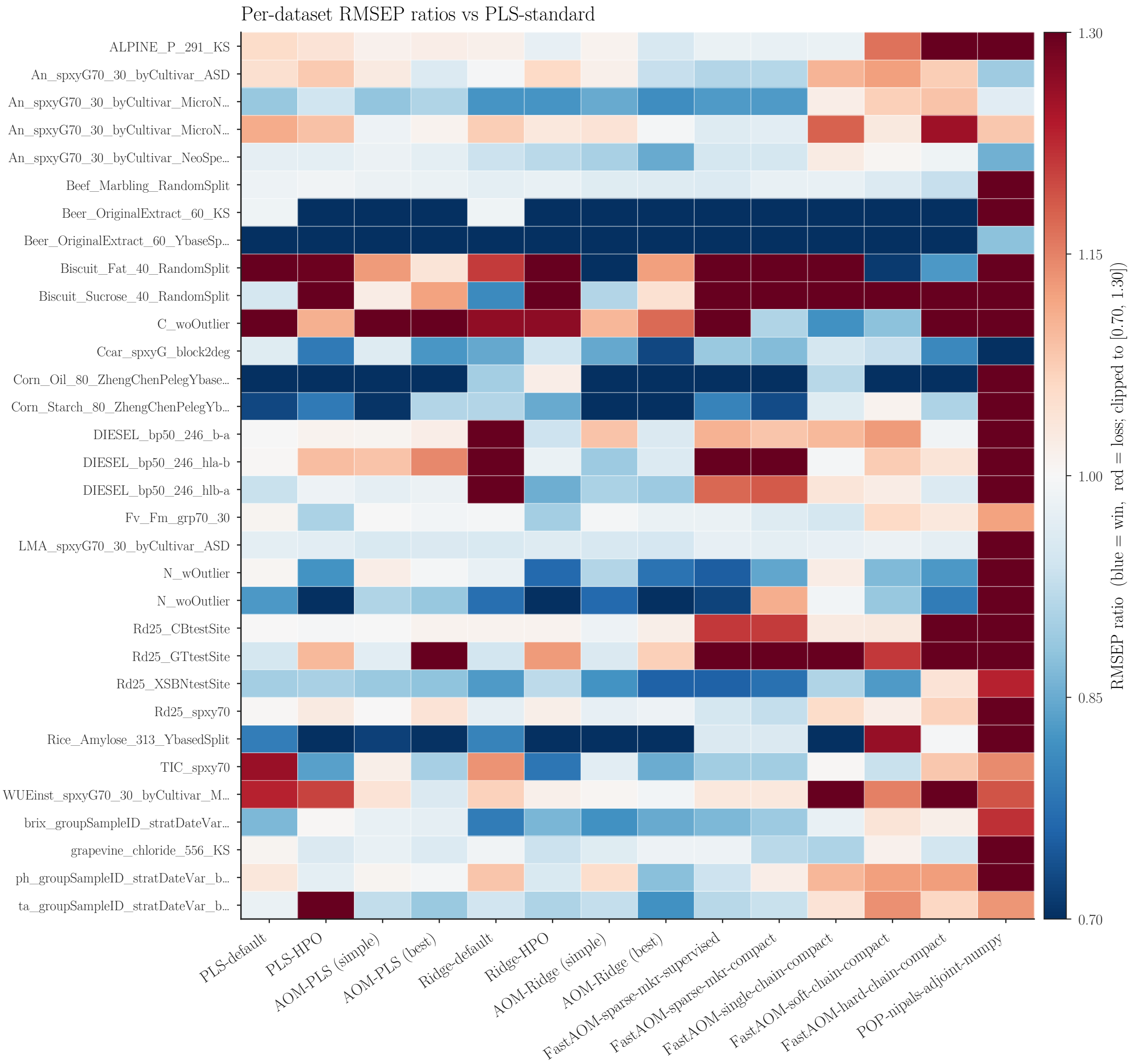

Figure 3: Per-dataset RMSEP ratios against PLS-standard. Columns are ordered by PLS, AOM-PLS, Ridge, AOM-Ridge and then exploratory FastAOM/POP variants; missing cells are shown in light gray.

Table 10: Per-dataset regression results on the strict intersection $N_\cap$ used in the paper. Relative RMSEP is divided by PLS-standard on the same dataset when the denominator is finite.

Dataset	Variant	RMSEP	Rel. RMSEP
ALPINE_P_291_KS	PLS-standard-numpy	0.05969	1.000
ALPINE_P_291_KS	AOM-compact-cv5-numpy	0.06079	1.018
ALPINE_P_291_KS	ASLS-AOM-compact-cv5-numpy	0.0606	1.015
ALPINE_P_291_KS	AOM-default-nipals-adjoint-numpy	0.06733	1.128
ALPINE_P_291_KS	nirs4all-AOM-PLS-default	0.06733	1.128
ALPINE_P_291_KS	FastAOM-sparse-mkr-supervised	0.05844	0.979
ALPINE_P_291_KS	FastAOM-sparse-mkr-compact	0.05843	0.979
ALPINE_P_291_KS	FastAOM-single-chain-compact	0.05846	0.979
ALPINE_P_291_KS	FastAOM-soft-chain-compact	0.06937	1.162
ALPINE_P_291_KS	FastAOM-hard-chain-compact	0.07823	1.311
ALPINE_P_291_KS	FastAOM-hard-chain-osc	0.07513	1.259
ALPINE_P_291_KS	FastAOM-hard-chain-asls	0.07621	1.277
ALPINE_P_291_KS	FastAOM-hard-chain-multibase	0.08579	1.437
ALPINE_P_291_KS	FastAOM-hard-chain-supervised	0.07941	1.330
ALPINE_P_291_KS	POP-nipals-adjoint-numpy	0.1007	1.687
ALPINE_P_291_KS	nirs4all-POP-PLS-default	0.4153	6.957
An_spxyG70_30_byCultivar_ASD	PLS-standard-numpy	3.868	1.000
An_spxyG70_30_byCultivar_ASD	AOM-compact-cv5-numpy	4.078	1.054
An_spxyG70_30_byCultivar_ASD	ASLS-AOM-compact-cv5-numpy	3.791	0.980
An_spxyG70_30_byCultivar_ASD	AOM-default-nipals-adjoint-numpy	3.904	1.009
An_spxyG70_30_byCultivar_ASD	nirs4all-AOM-PLS-default	3.904	1.009
An_spxyG70_30_byCultivar_ASD	FastAOM-sparse-mkr-supervised	3.524	0.911
An_spxyG70_30_byCultivar_ASD	FastAOM-sparse-mkr-compact	3.539	0.915
An_spxyG70_30_byCultivar_ASD	FastAOM-single-chain-compact	4.264	1.102
An_spxyG70_30_byCultivar_ASD	FastAOM-soft-chain-compact	4.339	1.122
An_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-compact	4.15	1.073
An_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-osc	4.788	1.238
An_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-asls	3.943	1.019
An_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-multibase	5.477	1.416
An_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-supervised	4.323	1.118
An_spxyG70_30_byCultivar_ASD	POP-nipals-adjoint-numpy	3.451	0.892

Dataset	Variant	RMSEP	Rel. RMSEP
An_spxyG70_30_byCultivar_ASD	nirs4all-POP-PLS-default	23.61	6.105
An_spxyG70_30_byCultivar_MicroNIR	PLS-standard-numpy	4.315	1.000
An_spxyG70_30_byCultivar_MicroNIR	AOM-compact-cv5-numpy	3.526	0.817
An_spxyG70_30_byCultivar_MicroNIR	ASLS-AOM-compact-cv5-numpy	3.622	0.840
An_spxyG70_30_byCultivar_MicroNIR	AOM-default-nipals-adjoint-numpy	4.343	1.006
An_spxyG70_30_byCultivar_MicroNIR	nirs4all-AOM-PLS-default	4.343	1.006
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-sparse-mkr-supervised	3.586	0.831
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-sparse-mkr-compact	3.586	0.831
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-single-chain-compact	4.404	1.021
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-soft-chain-compact	4.627	1.072
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-hard-chain-compact	4.685	1.086
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-hard-chain-osc	4.809	1.115
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-hard-chain-asls	5.084	1.178
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-hard-chain-multibase	4.947	1.146
An_spxyG70_30_byCultivar_MicroNIR	FastAOM-hard-chain-supervised	4.678	1.084
An_spxyG70_30_byCultivar_MicroNIR	POP-nipals-adjoint-numpy	4.166	0.966
An_spxyG70_30_byCultivar_MicroNIR	nirs4all-POP-PLS-default	21.22	4.919
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	PLS-standard-numpy	3.863	1.000
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	AOM-compact-cv5-numpy	3.791	0.981
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	ASLS-AOM-compact-cv5-numpy	3.618	0.936
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	AOM-default-nipals-adjoint-numpy	4.353	1.127
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	nirs4all-AOM-PLS-default	4.353	1.127
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-sparse-mkr-supervised	3.712	0.961
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-sparse-mkr-compact	3.74	0.968
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-single-chain-compact	4.549	1.178
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-soft-chain-compact	3.972	1.028

Dataset	Variant	RMSEP	Rel. RMSEP
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-compact	4.842	1.253
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-osc	4.295	1.112
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-asls	4.858	1.257
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-multibase	5.064	1.311
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-supervised	4.586	1.187
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	POP-nipals-adjoint-numpy	4.177	1.081
An_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	nirs4all-POP-PLS-default	22.44	5.808
An_spxyG70_30_byCultivar_NeoSpectra	PLS-standard-numpy	5.198	1.000
An_spxyG70_30_byCultivar_NeoSpectra	AOM-compact-cv5-numpy	5.156	0.992
An_spxyG70_30_byCultivar_NeoSpectra	ASLS-AOM-compact-cv5-numpy	4.978	0.958
An_spxyG70_30_byCultivar_NeoSpectra	AOM-default-nipals-adjoint-numpy	5.525	1.063
An_spxyG70_30_byCultivar_NeoSpectra	nirs4all-AOM-PLS-default	5.525	1.063
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-sparse-mkr-supervised	4.929	0.948
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-sparse-mkr-compact	4.929	0.948
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-single-chain-compact	5.322	1.024
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-soft-chain-compact	5.232	1.007
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-hard-chain-compact	5.125	0.986
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-hard-chain-osc	5.125	0.986
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-hard-chain-asls	5.223	1.005
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-hard-chain-multibase	5.277	1.015
An_spxyG70_30_byCultivar_NeoSpectra	FastAOM-hard-chain-supervised	5.174	0.996
An_spxyG70_30_byCultivar_NeoSpectra	POP-nipals-adjoint-numpy	4.452	0.857
An_spxyG70_30_byCultivar_NeoSpectra	nirs4all-POP-PLS-default	66.16	12.730
Beef_Marbling_RandomSplit	PLS-standard-numpy	74.82	1.000
Beef_Marbling_RandomSplit	AOM-compact-cv5-numpy	75.06	1.003
Beef_Marbling_RandomSplit	ASLS-AOM-compact-cv5-numpy	72.98	0.975
Beef_Marbling_RandomSplit	AOM-default-nipals-adjoint-numpy	72.78	0.973
Beef_Marbling_RandomSplit	nirs4all-AOM-PLS-default	72.78	0.973

Dataset	Variant	RMSEP	Rel. RMSEP
Beef_Marbling_RandomSplit	FastAOM-sparse-mkr-supervised	71.74	0.959
Beef_Marbling_RandomSplit	FastAOM-sparse-mkr-compact	73.08	0.977
Beef_Marbling_RandomSplit	FastAOM-single-chain-compact	73.12	0.977
Beef_Marbling_RandomSplit	FastAOM-soft-chain-compact	71.72	0.959
Beef_Marbling_RandomSplit	FastAOM-hard-chain-compact	69.62	0.930
Beef_Marbling_RandomSplit	FastAOM-hard-chain-osc	69.62	0.930
Beef_Marbling_RandomSplit	FastAOM-hard-chain-asls	71.2	0.952
Beef_Marbling_RandomSplit	FastAOM-hard-chain-multibase	71.82	0.960
Beef_Marbling_RandomSplit	FastAOM-hard-chain-supervised	72.7	0.972
Beef_Marbling_RandomSplit	POP-nipals-adjoint-numpy	98.66	1.319
Beef_Marbling_RandomSplit	nirs4all-POP-PLS-default	564.5	7.544
Beer_OriginalExtract_60_KS	PLS-standard-numpy	0.5784	1.000
Beer_OriginalExtract_60_KS	AOM-compact-cv5-numpy	0.3289	0.569
Beer_OriginalExtract_60_KS	ASLS-AOM-compact-cv5-numpy	0.2369	0.410
Beer_OriginalExtract_60_KS	AOM-default-nipals-adjoint-numpy	0.2042	0.353
Beer_OriginalExtract_60_KS	nirs4all-AOM-PLS-default	0.2042	0.353
Beer_OriginalExtract_60_KS	FastAOM-sparse-mkr-supervised	0.1611	0.279
Beer_OriginalExtract_60_KS	FastAOM-sparse-mkr-compact	0.1653	0.286
Beer_OriginalExtract_60_KS	FastAOM-single-chain-compact	0.2382	0.412
Beer_OriginalExtract_60_KS	FastAOM-soft-chain-compact	0.2384	0.412
Beer_OriginalExtract_60_KS	FastAOM-hard-chain-compact	0.245	0.424
Beer_OriginalExtract_60_KS	FastAOM-hard-chain-osc	0.2452	0.424
Beer_OriginalExtract_60_KS	FastAOM-hard-chain-asls	0.2431	0.420
Beer_OriginalExtract_60_KS	FastAOM-hard-chain-multibase	0.245	0.424
Beer_OriginalExtract_60_KS	FastAOM-hard-chain-supervised	0.2449	0.423
Beer_OriginalExtract_60_KS	POP-nipals-adjoint-numpy	1.624	2.809
Beer_OriginalExtract_60_KS	nirs4all-POP-PLS-default	0.9814	1.697
Beer_OriginalExtract_60_YbaseSplit	PLS-standard-numpy	0.9263	1.000
Beer_OriginalExtract_60_YbaseSplit	AOM-compact-cv5-numpy	0.2793	0.302
Beer_OriginalExtract_60_YbaseSplit	ASLS-AOM-compact-cv5-numpy	0.2538	0.274
Beer_OriginalExtract_60_YbaseSplit	AOM-default-nipals-adjoint-numpy	0.2107	0.227
Beer_OriginalExtract_60_YbaseSplit	nirs4all-AOM-PLS-default	0.2107	0.227
Beer_OriginalExtract_60_YbaseSplit	FastAOM-sparse-mkr-supervised	0.2766	0.299
Beer_OriginalExtract_60_YbaseSplit	FastAOM-sparse-mkr-compact	0.2713	0.293
Beer_OriginalExtract_60_YbaseSplit	FastAOM-single-chain-compact	0.3005	0.324

Dataset	Variant	RMSEP	Rel. RMSEP
Beer_OriginalExtract_60_YbaseSplit	FastAOM-soft-chain-compact	0.3082	0.333
Beer_OriginalExtract_60_YbaseSplit	FastAOM-hard-chain-compact	0.316	0.341
Beer_OriginalExtract_60_YbaseSplit	FastAOM-hard-chain-osc	0.3182	0.344
Beer_OriginalExtract_60_YbaseSplit	FastAOM-hard-chain-asls	0.3107	0.335
Beer_OriginalExtract_60_YbaseSplit	FastAOM-hard-chain-multibase	0.3094	0.334
Beer_OriginalExtract_60_YbaseSplit	FastAOM-hard-chain-supervised	0.3189	0.344
Beer_OriginalExtract_60_YbaseSplit	POP-nipals-adjoint-numpy	0.8123	0.877
Beer_OriginalExtract_60_YbaseSplit	nirs4all-POP-PLS-default	0.9859	1.064
Biscuit_Fat_40_RandomSplit	PLS-standard-numpy	0.4197	1.000
Biscuit_Fat_40_RandomSplit	AOM-compact-cv5-numpy	0.4421	1.054
Biscuit_Fat_40_RandomSplit	ASLS-AOM-compact-cv5-numpy	0.4371	1.042
Biscuit_Fat_40_RandomSplit	AOM-default-nipals-adjoint-numpy	0.5491	1.308
Biscuit_Fat_40_RandomSplit	nirs4all-AOM-PLS-default	0.5491	1.308
Biscuit_Fat_40_RandomSplit	FastAOM-sparse-mkr-supervised	0.5547	1.322
Biscuit_Fat_40_RandomSplit	FastAOM-sparse-mkr-compact	0.5547	1.322
Biscuit_Fat_40_RandomSplit	FastAOM-single-chain-compact	0.8011	1.909
Biscuit_Fat_40_RandomSplit	FastAOM-soft-chain-compact	0.2989	0.712
Biscuit_Fat_40_RandomSplit	FastAOM-hard-chain-compact	0.3477	0.828
Biscuit_Fat_40_RandomSplit	FastAOM-hard-chain-osc	0.5561	1.325
Biscuit_Fat_40_RandomSplit	FastAOM-hard-chain-asls	0.4015	0.957
Biscuit_Fat_40_RandomSplit	FastAOM-hard-chain-multibase	0.5485	1.307
Biscuit_Fat_40_RandomSplit	FastAOM-hard-chain-supervised	1.559	3.716
Biscuit_Fat_40_RandomSplit	POP-nipals-adjoint-numpy	2.035	4.850
Biscuit_Fat_40_RandomSplit	nirs4all-POP-PLS-default	5.867	13.979
Biscuit_Sucrose_40_RandomSplit	PLS-standard-numpy	1.154	1.000
Biscuit_Sucrose_40_RandomSplit	AOM-compact-cv5-numpy	1.165	1.009
Biscuit_Sucrose_40_RandomSplit	ASLS-AOM-compact-cv5-numpy	0.8817	0.764
Biscuit_Sucrose_40_RandomSplit	AOM-default-nipals-adjoint-numpy	4.347	3.765
Biscuit_Sucrose_40_RandomSplit	nirs4all-AOM-PLS-default	4.347	3.765
Biscuit_Sucrose_40_RandomSplit	FastAOM-sparse-mkr-supervised	4.516	3.912
Biscuit_Sucrose_40_RandomSplit	FastAOM-sparse-mkr-compact	4.035	3.495
Biscuit_Sucrose_40_RandomSplit	FastAOM-single-chain-compact	3.124	2.706
Biscuit_Sucrose_40_RandomSplit	FastAOM-soft-chain-compact	3.35	2.901
Biscuit_Sucrose_40_RandomSplit	FastAOM-hard-chain-compact	3.082	2.670
Biscuit_Sucrose_40_RandomSplit	FastAOM-hard-chain-osc	2.486	2.153

Dataset	Variant	RMSEP	Rel. RMSEP
Biscuit_Sucrose_40_RandomSplit	FastAOM-hard-chain-asls	3.29	2.850
Biscuit_Sucrose_40_RandomSplit	FastAOM-hard-chain-multibase	2.32	2.009
Biscuit_Sucrose_40_RandomSplit	FastAOM-hard-chain-supervised	2.779	2.407
Biscuit_Sucrose_40_RandomSplit	POP-nipals-adjoint-numpy	1.938	1.679
Biscuit_Sucrose_40_RandomSplit	nirs4all-POP-PLS-default	10.93	9.464
C_woOutlier	PLS-standard-numpy	1.602	1.000
C_woOutlier	AOM-compact-cv5-numpy	2.578	1.610
C_woOutlier	ASLS-AOM-compact-cv5-numpy	2.437	1.521
C_woOutlier	AOM-default-nipals-adjoint-numpy	2.328	1.453
C_woOutlier	nirs4all-AOM-PLS-default	2.328	1.453
C_woOutlier	FastAOM-sparse-mkr-supervised	2.494	1.557
C_woOutlier	FastAOM-sparse-mkr-compact	1.457	0.909
C_woOutlier	FastAOM-single-chain-compact	1.311	0.818
C_woOutlier	FastAOM-soft-chain-compact	1.406	0.878
C_woOutlier	FastAOM-hard-chain-compact	2.194	1.370
C_woOutlier	FastAOM-hard-chain-osc	1.41	0.880
C_woOutlier	FastAOM-hard-chain-asls	2.248	1.404
C_woOutlier	FastAOM-hard-chain-multibase	1.297	0.810
C_woOutlier	FastAOM-hard-chain-supervised	1.265	0.789
C_woOutlier	POP-nipals-adjoint-numpy	2.759	1.722
C_woOutlier	nirs4all-POP-PLS-default	9.058	5.655
Ccar_spxyG_block2deg	PLS-standard-numpy	70.98	1.000
Ccar_spxyG_block2deg	AOM-compact-cv5-numpy	70.98	1.000
Ccar_spxyG_block2deg	ASLS-AOM-compact-cv5-numpy	55.76	0.786
Ccar_spxyG_block2deg	AOM-default-nipals-adjoint-numpy	81.25	1.145
Ccar_spxyG_block2deg	nirs4all-AOM-PLS-default	81.25	1.145
Ccar_spxyG_block2deg	FastAOM-sparse-mkr-supervised	63.12	0.889
Ccar_spxyG_block2deg	FastAOM-sparse-mkr-compact	61.73	0.870
Ccar_spxyG_block2deg	FastAOM-single-chain-compact	67.3	0.948
Ccar_spxyG_block2deg	FastAOM-soft-chain-compact	66.13	0.932
Ccar_spxyG_block2deg	FastAOM-hard-chain-compact	57.33	0.808
Ccar_spxyG_block2deg	FastAOM-hard-chain-osc	52.94	0.746
Ccar_spxyG_block2deg	FastAOM-hard-chain-asls	61.44	0.866
Ccar_spxyG_block2deg	FastAOM-hard-chain-multibase	57.33	0.808
Ccar_spxyG_block2deg	FastAOM-hard-chain-supervised	61.98	0.873

Dataset	Variant	RMSEP	Rel. RMSEP
Ccar_spxyG_block2deg	POP-nipals-adjoint-numpy	39.16	0.552
Ccar_spxyG_block2deg	nirs4all-POP-PLS-default	160.6	2.262
Corn_Oil_80_ZhengChenPelegYbaseSplit	PLS-standard-numpy	0.06628	1.000
Corn_Oil_80_ZhengChenPelegYbaseSplit	AOM-compact-cv5-numpy	0.02652	0.400
Corn_Oil_80_ZhengChenPelegYbaseSplit	ASLS-AOM-compact-cv5-numpy	0.0237	0.358
Corn_Oil_80_ZhengChenPelegYbaseSplit	AOM-default-nipals-adjoint-numpy	0.02218	0.335
Corn_Oil_80_ZhengChenPelegYbaseSplit	nirs4all-AOM-PLS-default	0.02218	0.335
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-sparse-mkr-supervised	0.02497	0.377
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-sparse-mkr-compact	0.02956	0.446
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-single-chain-compact	0.06081	0.918
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-soft-chain-compact	0.04219	0.637
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-compact	0.04373	0.660
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-osc	0.04373	0.660
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-asls	0.03496	0.527
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-multibase	0.02996	0.452
Corn_Oil_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-supervised	0.04208	0.635
Corn_Oil_80_ZhengChenPelegYbaseSplit	POP-nipals-adjoint-numpy	0.1604	2.420
Corn_Oil_80_ZhengChenPelegYbaseSplit	nirs4all-POP-PLS-default	0.5305	8.004
Corn_Starch_80_ZhengChenPelegYbaseSplit	PLS-standard-numpy	0.1972	1.000
Corn_Starch_80_ZhengChenPelegYbaseSplit	AOM-compact-cv5-numpy	0.1394	0.707
Corn_Starch_80_ZhengChenPelegYbaseSplit	ASLS-AOM-compact-cv5-numpy	0.1818	0.922
Corn_Starch_80_ZhengChenPelegYbaseSplit	AOM-default-nipals-adjoint-numpy	0.1695	0.859
Corn_Starch_80_ZhengChenPelegYbaseSplit	nirs4all-AOM-PLS-default	0.1695	0.859
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-sparse-mkr-supervised	0.1577	0.799
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-sparse-mkr-compact	0.1443	0.732
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-single-chain-compact	0.1902	0.965
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-soft-chain-compact	0.1993	1.010
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-compact	0.179	0.907
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-osc	0.1829	0.927
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-asls	0.2488	1.261
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-multibase	0.1671	0.848
Corn_Starch_80_ZhengChenPelegYbaseSplit	FastAOM-hard-chain-supervised	0.1344	0.682
Corn_Starch_80_ZhengChenPelegYbaseSplit	POP-nipals-adjoint-numpy	0.7876	3.994
Corn_Starch_80_ZhengChenPelegYbaseSplit	nirs4all-POP-PLS-default	4.776	24.217
DIESEL_bp50_246_b-a	PLS-standard-numpy	3.154	1.000

Dataset	Variant	RMSEP	Rel. RMSEP
DIESEL_bp50_246_b-a	AOM-compact-cv5-numpy	3.21	1.018
DIESEL_bp50_246_b-a	ASLS-AOM-compact-cv5-numpy	3.225	1.023
DIESEL_bp50_246_b-a	AOM-default-nipals-adjoint-numpy	3.239	1.027
DIESEL_bp50_246_b-a	nirs4all-AOM-PLS-default	3.239	1.027
DIESEL_bp50_246_b-a	FastAOM-sparse-mkr-supervised	3.483	1.105
DIESEL_bp50_246_b-a	FastAOM-sparse-mkr-compact	3.413	1.082
DIESEL_bp50_246_b-a	FastAOM-single-chain-compact	3.459	1.097
DIESEL_bp50_246_b-a	FastAOM-soft-chain-compact	3.558	1.128
DIESEL_bp50_246_b-a	FastAOM-hard-chain-compact	3.122	0.990
DIESEL_bp50_246_b-a	FastAOM-hard-chain-osc	3.166	1.004
DIESEL_bp50_246_b-a	FastAOM-hard-chain-asls	3.169	1.005
DIESEL_bp50_246_b-a	FastAOM-hard-chain-multibase	3.191	1.012
DIESEL_bp50_246_b-a	FastAOM-hard-chain-supervised	3.419	1.084
DIESEL_bp50_246_b-a	POP-nipals-adjoint-numpy	11.52	3.654
DIESEL_bp50_246_b-a	nirs4all-POP-PLS-default	62.26	19.743
DIESEL_bp50_246_hla-b	PLS-standard-numpy	2.794	1.000
DIESEL_bp50_246_hla-b	AOM-compact-cv5-numpy	3.104	1.111
DIESEL_bp50_246_hla-b	ASLS-AOM-compact-cv5-numpy	3.114	1.114
DIESEL_bp50_246_hla-b	AOM-default-nipals-adjoint-numpy	2.73	0.977
DIESEL_bp50_246_hla-b	nirs4all-AOM-PLS-default	2.73	0.977
DIESEL_bp50_246_hla-b	FastAOM-sparse-mkr-supervised	3.774	1.350
DIESEL_bp50_246_hla-b	FastAOM-sparse-mkr-compact	3.987	1.427
DIESEL_bp50_246_hla-b	FastAOM-single-chain-compact	2.776	0.993
DIESEL_bp50_246_hla-b	FastAOM-soft-chain-compact	3.008	1.076
DIESEL_bp50_246_hla-b	FastAOM-hard-chain-compact	2.902	1.038
DIESEL_bp50_246_hla-b	FastAOM-hard-chain-osc	2.995	1.072
DIESEL_bp50_246_hla-b	FastAOM-hard-chain-asls	2.979	1.066
DIESEL_bp50_246_hla-b	FastAOM-hard-chain-multibase	3.079	1.102
DIESEL_bp50_246_hla-b	FastAOM-hard-chain-supervised	2.953	1.057
DIESEL_bp50_246_hla-b	POP-nipals-adjoint-numpy	16.78	6.006
DIESEL_bp50_246_hla-b	nirs4all-POP-PLS-default	41.51	14.854
DIESEL_bp50_246_hlb-a	PLS-standard-numpy	3.137	1.000
DIESEL_bp50_246_hlb-a	AOM-compact-cv5-numpy	3.104	0.989
DIESEL_bp50_246_hlb-a	ASLS-AOM-compact-cv5-numpy	3.14	1.001
DIESEL_bp50_246_hlb-a	AOM-default-nipals-adjoint-numpy	3.703	1.180

Dataset	Variant	RMSEP	Rel. RMSEP
DIESEL__bp50__246__hlb-a	nirs4all-AOM-PLS-default	3.703	1.180
DIESEL__bp50__246__hlb-a	FastAOM-sparse-mkr-supervised	3.675	1.171
DIESEL__bp50__246__hlb-a	FastAOM-sparse-mkr-compact	3.714	1.184
DIESEL__bp50__246__hlb-a	FastAOM-single-chain-compact	3.25	1.036
DIESEL__bp50__246__hlb-a	FastAOM-soft-chain-compact	3.205	1.022
DIESEL__bp50__246__hlb-a	FastAOM-hard-chain-compact	3.005	0.958
DIESEL__bp50__246__hlb-a	FastAOM-hard-chain-osc	3.047	0.971
DIESEL__bp50__246__hlb-a	FastAOM-hard-chain-asls	2.978	0.949
DIESEL__bp50__246__hlb-a	FastAOM-hard-chain-multibase	3.048	0.972
DIESEL__bp50__246__hlb-a	FastAOM-hard-chain-supervised	2.902	0.925
DIESEL__bp50__246__hlb-a	POP-nipals-adjoint-numpy	14.34	4.570
DIESEL__bp50__246__hlb-a	nirs4all-POP-PLS-default	58.51	18.650
Fv__Fm__grp70__30	PLS-standard-numpy	0.03124	1.000
Fv__Fm__grp70__30	AOM-compact-cv5-numpy	0.0313	1.002
Fv__Fm__grp70__30	ASLS-AOM-compact-cv5-numpy	0.03072	0.983
Fv__Fm__grp70__30	AOM-default-nipals-adjoint-numpy	0.03328	1.065
Fv__Fm__grp70__30	nirs4all-AOM-PLS-default	0.03328	1.065
Fv__Fm__grp70__30	FastAOM-sparse-mkr-supervised	0.03061	0.980
Fv__Fm__grp70__30	FastAOM-sparse-mkr-compact	0.03005	0.962
Fv__Fm__grp70__30	FastAOM-single-chain-compact	0.02958	0.947
Fv__Fm__grp70__30	FastAOM-soft-chain-compact	0.03314	1.061
Fv__Fm__grp70__30	FastAOM-hard-chain-compact	0.03223	1.032
Fv__Fm__grp70__30	FastAOM-hard-chain-osc	0.03206	1.026
Fv__Fm__grp70__30	FastAOM-hard-chain-asls	0.03578	1.145
Fv__Fm__grp70__30	FastAOM-hard-chain-multibase	0.03098	0.992
Fv__Fm__grp70__30	FastAOM-hard-chain-supervised	0.03294	1.054
Fv__Fm__grp70__30	POP-nipals-adjoint-numpy	0.03504	1.122
Fv__Fm__grp70__30	nirs4all-POP-PLS-default	0.1362	4.358
LMA__spxyG70__30__byCultivar__ASD	PLS-standard-numpy	0.3177	1.000
LMA__spxyG70__30__byCultivar__ASD	AOM-compact-cv5-numpy	0.3032	0.954
LMA__spxyG70__30__byCultivar__ASD	ASLS-AOM-compact-cv5-numpy	0.3042	0.958
LMA__spxyG70__30__byCultivar__ASD	AOM-default-nipals-adjoint-numpy	0.3099	0.976
LMA__spxyG70__30__byCultivar__ASD	nirs4all-AOM-PLS-default	0.3099	0.976
LMA__spxyG70__30__byCultivar__ASD	FastAOM-sparse-mkr-supervised	0.3099	0.976
LMA__spxyG70__30__byCultivar__ASD	FastAOM-sparse-mkr-compact	0.3082	0.970

Dataset	Variant	RMSEP	Rel. RMSEP
LMA_spxyG70_30_byCultivar_ASD	FastAOM-single-chain-compact	0.3102	0.977
LMA_spxyG70_30_byCultivar_ASD	FastAOM-soft-chain-compact	0.3123	0.983
LMA_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-compact	0.3094	0.974
LMA_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-osc	0.307	0.966
LMA_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-asls	0.3168	0.997
LMA_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-multibase	0.3086	0.971
LMA_spxyG70_30_byCultivar_ASD	FastAOM-hard-chain-supervised	0.3019	0.950
LMA_spxyG70_30_byCultivar_ASD	POP-nipals-adjoint-numpy	0.7741	2.437
LMA_spxyG70_30_byCultivar_ASD	nirs4all-POP-PLS-default	1.123	3.535
N_wOutlier	PLS-standard-numpy	0.3511	1.000
N_wOutlier	AOM-compact-cv5-numpy	0.3577	1.019
N_wOutlier	ASLS-AOM-compact-cv5-numpy	0.3495	0.995
N_wOutlier	AOM-default-nipals-adjoint-numpy	0.3481	0.991
N_wOutlier	nirs4all-AOM-PLS-default	0.3481	0.991
N_wOutlier	FastAOM-sparse-mkr-supervised	0.2643	0.753
N_wOutlier	FastAOM-sparse-mkr-compact	0.2967	0.845
N_wOutlier	FastAOM-single-chain-compact	0.3589	1.022
N_wOutlier	FastAOM-soft-chain-compact	0.3046	0.867
N_wOutlier	FastAOM-hard-chain-compact	0.2908	0.828
N_wOutlier	FastAOM-hard-chain-osc	0.2867	0.817
N_wOutlier	FastAOM-hard-chain-asls	0.2848	0.811
N_wOutlier	FastAOM-hard-chain-multibase	0.2995	0.853
N_wOutlier	FastAOM-hard-chain-supervised	0.3038	0.865
N_wOutlier	POP-nipals-adjoint-numpy	1.392	3.965
N_wOutlier	nirs4all-POP-PLS-default	2.186	6.227
N_woOutlier	PLS-standard-numpy	0.3538	1.000
N_woOutlier	AOM-compact-cv5-numpy	0.3223	0.911
N_woOutlier	ASLS-AOM-compact-cv5-numpy	0.3134	0.886
N_woOutlier	AOM-default-nipals-adjoint-numpy	0.3394	0.959
N_woOutlier	nirs4all-AOM-PLS-default	0.3394	0.959
N_woOutlier	FastAOM-sparse-mkr-supervised	0.2559	0.723
N_woOutlier	FastAOM-sparse-mkr-compact	0.392	1.108
N_woOutlier	FastAOM-single-chain-compact	0.3512	0.993
N_woOutlier	FastAOM-soft-chain-compact	0.3135	0.886
N_woOutlier	FastAOM-hard-chain-compact	0.2792	0.789

Dataset	Variant	RMSEP	Rel. RMSEP
N_woOutlier	FastAOM-hard-chain-osc	0.2805	0.793
N_woOutlier	FastAOM-hard-chain-asls	0.2869	0.811
N_woOutlier	FastAOM-hard-chain-multibase	0.2865	0.810
N_woOutlier	FastAOM-hard-chain-supervised	0.2881	0.814
N_woOutlier	POP-nipals-adjoint-numpy	1.395	3.945
N_woOutlier	nirs4all-POP-PLS-default	2.237	6.324
Rd25_CBtestSite	PLS-standard-numpy	0.2288	1.000
Rd25_CBtestSite	AOM-compact-cv5-numpy	0.2288	1.000
Rd25_CBtestSite	ASLS-AOM-compact-cv5-numpy	0.2283	0.998
Rd25_CBtestSite	AOM-default-nipals-adjoint-numpy	0.2472	1.081
Rd25_CBtestSite	nirs4all-AOM-PLS-default	0.2472	1.081
Rd25_CBtestSite	FastAOM-sparse-mkr-supervised	0.2775	1.213
Rd25_CBtestSite	FastAOM-sparse-mkr-compact	0.2765	1.209
Rd25_CBtestSite	FastAOM-single-chain-compact	0.2348	1.026
Rd25_CBtestSite	FastAOM-soft-chain-compact	0.2355	1.030
Rd25_CBtestSite	FastAOM-hard-chain-compact	0.3009	1.315
Rd25_CBtestSite	FastAOM-hard-chain-osc	0.279	1.220
Rd25_CBtestSite	FastAOM-hard-chain-asls	0.2896	1.266
Rd25_CBtestSite	FastAOM-hard-chain-multibase	0.2859	1.250
Rd25_CBtestSite	FastAOM-hard-chain-supervised	0.2977	1.301
Rd25_CBtestSite	POP-nipals-adjoint-numpy	0.3628	1.586
Rd25_CBtestSite	nirs4all-POP-PLS-default	0.7211	3.152
Rd25_GTtestSite	PLS-standard-numpy	0.2042	1.000
Rd25_GTtestSite	AOM-compact-cv5-numpy	0.2086	1.022
Rd25_GTtestSite	ASLS-AOM-compact-cv5-numpy	0.3195	1.565
Rd25_GTtestSite	AOM-default-nipals-adjoint-numpy	0.204	0.999
Rd25_GTtestSite	nirs4all-AOM-PLS-default	0.204	0.999
Rd25_GTtestSite	FastAOM-sparse-mkr-supervised	0.3567	1.747
Rd25_GTtestSite	FastAOM-sparse-mkr-compact	0.3567	1.747
Rd25_GTtestSite	FastAOM-single-chain-compact	0.2757	1.351
Rd25_GTtestSite	FastAOM-soft-chain-compact	0.2477	1.213
Rd25_GTtestSite	FastAOM-hard-chain-compact	0.3302	1.618
Rd25_GTtestSite	FastAOM-hard-chain-osc	0.347	1.699
Rd25_GTtestSite	FastAOM-hard-chain-asls	0.2984	1.462
Rd25_GTtestSite	FastAOM-hard-chain-multibase	0.395	1.935

Dataset	Variant	RMSEP	Rel. RMSEP
Rd25_GTtestSite	FastAOM-hard-chain-supervised	0.2985	1.462
Rd25_GTtestSite	POP-nipals-adjoint-numpy	0.4259	2.086
Rd25_GTtestSite	nirs4all-POP-PLS-default	0.5902	2.891
Rd25_XSBNtestSite	PLS-standard-numpy	0.3158	1.000
Rd25_XSBNtestSite	AOM-compact-cv5-numpy	0.2847	0.902
Rd25_XSBNtestSite	ASLS-AOM-compact-cv5-numpy	0.2657	0.841
Rd25_XSBNtestSite	AOM-default-nipals-adjoint-numpy	0.2637	0.835
Rd25_XSBNtestSite	nirs4all-AOM-PLS-default	0.2637	0.835
Rd25_XSBNtestSite	FastAOM-sparse-mkr-supervised	0.2395	0.758
Rd25_XSBNtestSite	FastAOM-sparse-mkr-compact	0.2448	0.775
Rd25_XSBNtestSite	FastAOM-single-chain-compact	0.2869	0.909
Rd25_XSBNtestSite	FastAOM-soft-chain-compact	0.2618	0.829
Rd25_XSBNtestSite	FastAOM-hard-chain-compact	0.3291	1.042
Rd25_XSBNtestSite	FastAOM-hard-chain-osc	0.3295	1.043
Rd25_XSBNtestSite	FastAOM-hard-chain-asls	0.2804	0.888
Rd25_XSBNtestSite	FastAOM-hard-chain-multibase	0.303	0.960
Rd25_XSBNtestSite	FastAOM-hard-chain-supervised	0.2914	0.923
Rd25_XSBNtestSite	POP-nipals-adjoint-numpy	0.3887	1.231
Rd25_XSBNtestSite	nirs4all-POP-PLS-default	1.467	4.647
Rd25_spxy70	PLS-standard-numpy	0.1786	1.000
Rd25_spxy70	AOM-compact-cv5-numpy	0.1781	0.997
Rd25_spxy70	ASLS-AOM-compact-cv5-numpy	0.1863	1.043
Rd25_spxy70	AOM-default-nipals-adjoint-numpy	0.1802	1.009
Rd25_spxy70	nirs4all-AOM-PLS-default	0.1802	1.009
Rd25_spxy70	FastAOM-sparse-mkr-supervised	0.169	0.947
Rd25_spxy70	FastAOM-sparse-mkr-compact	0.1657	0.928
Rd25_spxy70	FastAOM-single-chain-compact	0.1883	1.055
Rd25_spxy70	FastAOM-soft-chain-compact	0.1822	1.020
Rd25_spxy70	FastAOM-hard-chain-compact	0.1909	1.069
Rd25_spxy70	FastAOM-hard-chain-osc	0.1934	1.083
Rd25_spxy70	FastAOM-hard-chain-asls	0.1802	1.009
Rd25_spxy70	FastAOM-hard-chain-multibase	0.1854	1.038
Rd25_spxy70	FastAOM-hard-chain-supervised	0.1805	1.011
Rd25_spxy70	POP-nipals-adjoint-numpy	0.2584	1.447
Rd25_spxy70	nirs4all-POP-PLS-default	0.6475	3.626

Dataset	Variant	RMSEP	Rel. RMSEP
Rice_Amylose_313_YbasedSplit	PLS-standard-numpy	3.245	1.000
Rice_Amylose_313_YbasedSplit	AOM-compact-cv5-numpy	2.471	0.762
Rice_Amylose_313_YbasedSplit	ASLS-AOM-compact-cv5-numpy	2.277	0.702
Rice_Amylose_313_YbasedSplit	AOM-default-nipals-adjoint-numpy	1.887	0.582
Rice_Amylose_313_YbasedSplit	nirs4all-AOM-PLS-default	1.887	0.582
Rice_Amylose_313_YbasedSplit	FastAOM-sparse-mkr-supervised	3.106	0.957
Rice_Amylose_313_YbasedSplit	FastAOM-sparse-mkr-compact	3.106	0.957
Rice_Amylose_313_YbasedSplit	FastAOM-single-chain-compact	2.202	0.679
Rice_Amylose_313_YbasedSplit	FastAOM-soft-chain-compact	4.095	1.262
Rice_Amylose_313_YbasedSplit	FastAOM-hard-chain-compact	3.234	0.997
Rice_Amylose_313_YbasedSplit	FastAOM-hard-chain-osc	2.935	0.905
Rice_Amylose_313_YbasedSplit	FastAOM-hard-chain-asls	3.143	0.969
Rice_Amylose_313_YbasedSplit	FastAOM-hard-chain-multibase	3.632	1.119
Rice_Amylose_313_YbasedSplit	FastAOM-hard-chain-supervised	3.221	0.993
Rice_Amylose_313_YbasedSplit	POP-nipals-adjoint-numpy	5.324	1.641
Rice_Amylose_313_YbasedSplit	nirs4all-POP-PLS-default	39.15	12.067
TIC_spxy70	PLS-standard-numpy	3.991	1.000
TIC_spxy70	AOM-compact-cv5-numpy	3.713	0.930
TIC_spxy70	ASLS-AOM-compact-cv5-numpy	3.698	0.926
TIC_spxy70	AOM-default-nipals-adjoint-numpy	3.092	0.775
TIC_spxy70	nirs4all-AOM-PLS-default	3.092	0.775
TIC_spxy70	FastAOM-sparse-mkr-supervised	3.577	0.896
TIC_spxy70	FastAOM-sparse-mkr-compact	3.577	0.896
TIC_spxy70	FastAOM-single-chain-compact	4.009	1.004
TIC_spxy70	FastAOM-soft-chain-compact	3.728	0.934
TIC_spxy70	FastAOM-hard-chain-compact	4.317	1.082
TIC_spxy70	FastAOM-hard-chain-osc	4.271	1.070
TIC_spxy70	FastAOM-hard-chain-asls	4.534	1.136
TIC_spxy70	FastAOM-hard-chain-multibase	4.253	1.066
TIC_spxy70	FastAOM-hard-chain-supervised	4.17	1.045
TIC_spxy70	POP-nipals-adjoint-numpy	4.557	1.142
TIC_spxy70	nirs4all-POP-PLS-default	9.156	2.294
WUEinst_spxyG70_30_byCultivar_MicroNIR_ NeoSpectra	PLS-standard-numpy	1.458	1.000

Dataset	Variant	RMSEP	Rel. RMSEP
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	AOM-compact-cv5-numpy	1.416	0.971
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	ASLS-AOM-compact-cv5-numpy	1.311	0.900
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	AOM-default-nipals-adjoint-numpy	1.556	1.067
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	nirs4all-AOM-PLS-default	1.556	1.067
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-sparse-mkr-supervised	1.503	1.031
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-sparse-mkr-compact	1.503	1.031
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-single-chain-compact	2.063	1.415
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-soft-chain-compact	1.678	1.151
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-compact	2.297	1.576
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-osc	2.297	1.576
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-asls	2.106	1.445
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-multibase	2.231	1.531
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	FastAOM-hard-chain-supervised	2.01	1.379
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	POP-nipals-adjoint-numpy	1.734	1.190
WUEinst_spxyG70_30_byCultivar_MicroNIR_NeoSpectra	nirs4all-POP-PLS-default	11.42	7.835
brix_groupSampleID_stratDateVar_balRows	PLS-standard-numpy	4.428	1.000
brix_groupSampleID_stratDateVar_balRows	AOM-compact-cv5-numpy	4.282	0.967
brix_groupSampleID_stratDateVar_balRows	ASLS-AOM-compact-cv5-numpy	4.213	0.952
brix_groupSampleID_stratDateVar_balRows	AOM-default-nipals-adjoint-numpy	4.163	0.940
brix_groupSampleID_stratDateVar_balRows	nirs4all-AOM-PLS-default	4.163	0.940

Dataset	Variant	RMSEP	Rel. RMSEP
brix__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-supervised	3.831	0.865
brix__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-compact	3.944	0.891
brix__groupSampleID__stratDateVar__balRows	FastAOM-single-chain-compact	4.334	0.979
brix__groupSampleID__stratDateVar__balRows	FastAOM-soft-chain-compact	4.6	1.039
brix__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-compact	4.509	1.018
brix__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-osc	4.685	1.058
brix__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-asls	4.533	1.024
brix__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-multibase	4.602	1.039
brix__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-supervised	4.463	1.008
brix__groupSampleID__stratDateVar__balRows	POP-nipals-adjoint-numpy	5.401	1.220
brix__groupSampleID__stratDateVar__balRows	nirs4all-POP-PLS-default	16.28	3.677
grapevine__chloride__556__KS	PLS-standard-numpy	1000	1.000
grapevine__chloride__556__KS	AOM-compact-cv5-numpy	977.5	0.978
grapevine__chloride__556__KS	ASLS-AOM-compact-cv5-numpy	946.4	0.946
grapevine__chloride__556__KS	AOM-default-nipals-adjoint-numpy	962.7	0.963
grapevine__chloride__556__KS	nirs4all-AOM-PLS-default	962.7	0.963
grapevine__chloride__556__KS	FastAOM-sparse-mkr-supervised	984.2	0.984
grapevine__chloride__556__KS	FastAOM-sparse-mkr-compact	918.7	0.919
grapevine__chloride__556__KS	FastAOM-single-chain-compact	906.5	0.907
grapevine__chloride__556__KS	FastAOM-soft-chain-compact	1013	1.013
grapevine__chloride__556__KS	FastAOM-hard-chain-compact	945.1	0.945
grapevine__chloride__556__KS	FastAOM-hard-chain-osc	963.3	0.963
grapevine__chloride__556__KS	FastAOM-hard-chain-asls	970.3	0.970
grapevine__chloride__556__KS	FastAOM-hard-chain-multibase	987.3	0.987
grapevine__chloride__556__KS	FastAOM-hard-chain-supervised	972.5	0.973
grapevine__chloride__556__KS	POP-nipals-adjoint-numpy	1482	1.482
grapevine__chloride__556__KS	nirs4all-POP-PLS-default	3774	3.775
ph__groupSampleID__stratDateVar__balRows	PLS-standard-numpy	0.3405	1.000
ph__groupSampleID__stratDateVar__balRows	AOM-compact-cv5-numpy	0.3405	1.000
ph__groupSampleID__stratDateVar__balRows	ASLS-AOM-compact-cv5-numpy	0.3385	0.994
ph__groupSampleID__stratDateVar__balRows	AOM-default-nipals-adjoint-numpy	0.3513	1.032
ph__groupSampleID__stratDateVar__balRows	nirs4all-AOM-PLS-default	0.3513	1.032
ph__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-supervised	0.3191	0.937
ph__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-compact	0.3472	1.020
ph__groupSampleID__stratDateVar__balRows	FastAOM-single-chain-compact	0.3746	1.100

Dataset	Variant	RMSEP	Rel. RMSEP
ph__groupSampleID__stratDateVar__balRows	FastAOM-soft-chain-compact	0.3824	1.123
ph__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-compact	0.3833	1.126
ph__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-osc	0.3681	1.081
ph__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-asls	0.3494	1.026
ph__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-multibase	0.3833	1.126
ph__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-supervised	0.3407	1.001
ph__groupSampleID__stratDateVar__balRows	POP-nipals-adjoint-numpy	0.4608	1.353
ph__groupSampleID__stratDateVar__balRows	nirs4all-POP-PLS-default	1.301	3.822
ta__groupSampleID__stratDateVar__balRows	PLS-standard-numpy	2.179	1.000
ta__groupSampleID__stratDateVar__balRows	AOM-compact-cv5-numpy	2.176	0.999
ta__groupSampleID__stratDateVar__balRows	ASLS-AOM-compact-cv5-numpy	1.936	0.889
ta__groupSampleID__stratDateVar__balRows	AOM-default-nipals-adjoint-numpy	1.873	0.860
ta__groupSampleID__stratDateVar__balRows	nirs4all-AOM-PLS-default	1.873	0.860
ta__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-supervised	1.996	0.916
ta__groupSampleID__stratDateVar__balRows	FastAOM-sparse-mkr-compact	2.033	0.933
ta__groupSampleID__stratDateVar__balRows	FastAOM-single-chain-compact	2.27	1.042
ta__groupSampleID__stratDateVar__balRows	FastAOM-soft-chain-compact	2.478	1.137
ta__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-compact	2.313	1.061
ta__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-osc	2.29	1.051
ta__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-asls	2.127	0.976
ta__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-multibase	2.313	1.061
ta__groupSampleID__stratDateVar__balRows	FastAOM-hard-chain-supervised	2.269	1.042
ta__groupSampleID__stratDateVar__balRows	POP-nipals-adjoint-numpy	2.465	1.132
ta__groupSampleID__stratDateVar__balRows	nirs4all-POP-PLS-default	8.063	3.701

9 Per-dataset prediction quality

Figure 4 shows the per-dataset RMSEP reduction for the selected AOM-PLS and AOM-Ridge variants against default and HPO references.

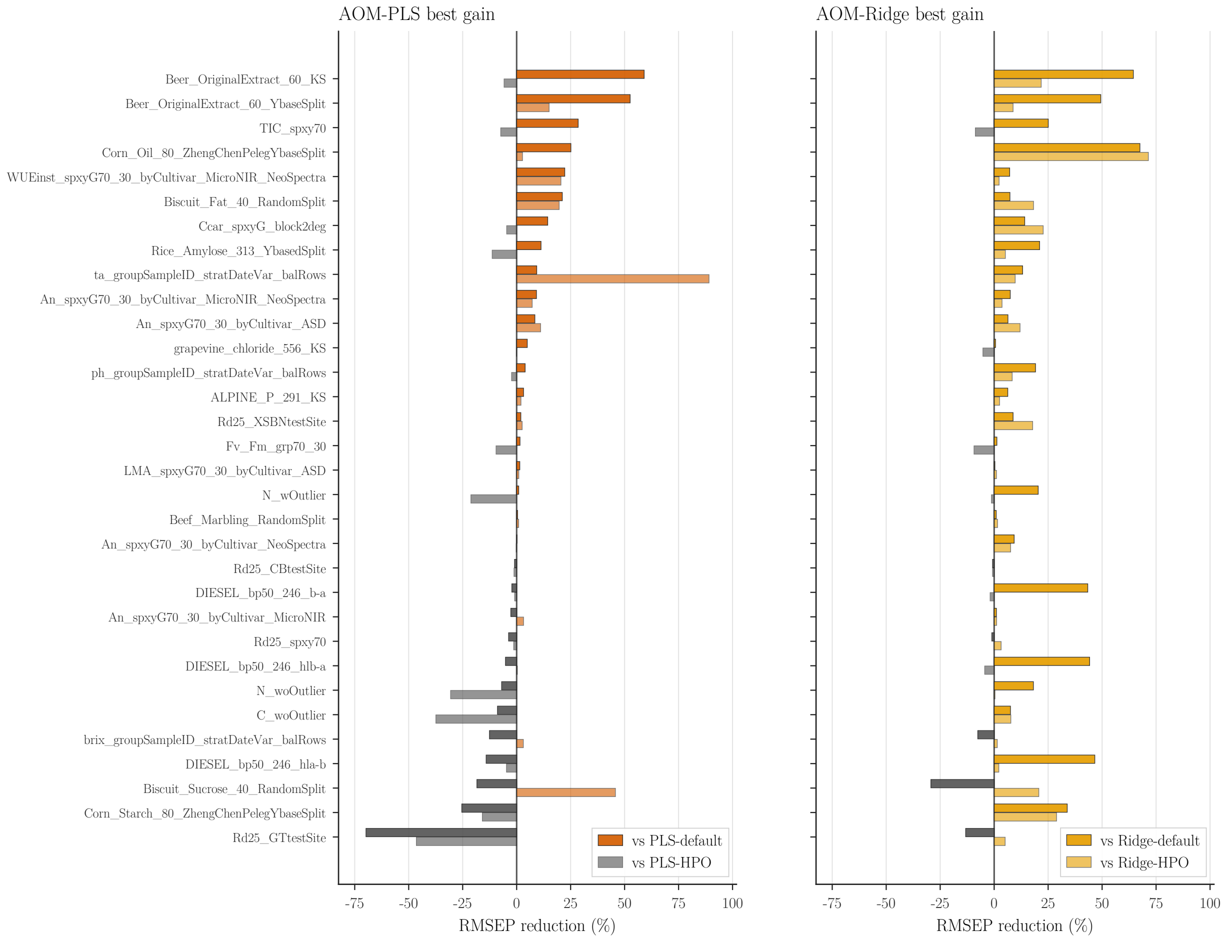

Figure 4: Per-dataset RMSEP reduction for selected AOM variants. Positive bars indicate lower RMSEP than the named reference; gray bars indicate a higher RMSEP.

10 Failure modes

Table 11 reports the datasets where the AOM-PLS family does not produce a clear improvement. Two manifest rows (`FinalScore_grp70_30_scoreQ` and `Tleaf_grp70_30`) cannot be fit by any of the nine logged variants because the source spectra contain NaN entries; they are kept here for transparency rather than as a comparison. The remaining rows are datasets where at least one AOM-PLS variant has elevated RMSEP relative to the headline figures in the main text: rice amylose, brix on the strawberry-puree split with sample-ID grouping, the colza C dataset and the seven dataset block from `N_*` and the `Brix_spxy70` berry split. These are the datasets that drive the selected ASLS-AOM-compact-cv5 paired ratio away from its overall median, and they are where a domain expert would expect SNV, MSC or a fitted baseline correction to play a larger role than a strict linear bank can. POP-PLS variants are particularly weak on this subset (RMSEPs in the range 1.39–5.32 on rows where the linear references stay below 2.0), confirming the main-text observation that per-component preprocessing flexibility hurts more than it helps when the calibration set is small. The per-dataset gain figure (Section 9) shows the same picture as a bar chart.

Table 11: Datasets where at least one AOM-PLS variant has elevated RMSEP or where every variant fails because of NaN inputs. "Variants logged" counts the nine AOM-PLS and POP-PLS rows checked; "Finite RMSEP rows" counts how many produced a finite test RMSEP.

Dataset	Variants logged	Finite RMSEP rows	Min RMSEP	Max RMSEP
FinalScore_grp70_30_scoreQ	9	0	n/a	n/a
Tleaf_grp70_30	9	0	n/a	n/a
Brix_spxy70	9	9	0.9499	2.0182
C_woOutlier	9	9	1.6018	2.7590
LMA_spxyG70_30_byCultivar_ASD	9	9	0.3032	0.7767
N_wOutlier	9	9	0.3429	1.3923
N_woOutlier	9	9	0.3134	1.3955
Rice_Amylose_313_YbasedSplit	9	9	1.8873	5.3238
brix_groupSampleID_stratDateVar_balRows	9	9	4.1629	5.3942

11 Classification Full Results

Table 12: Full classification paired comparisons. Positive differences favour the AOM or POP candidate over PLS-DA.

Comparison	N	Median Δ balanced acc.	95% CI	Wins; p_{Holm}
AOM-PLS-DA-global-simpls-covariance vs PLS-DA	13	0.159	0.129–0.422	12/13; 0.007
POP-PLS-DA-simpls-covariance vs PLS-DA	13	0.052	0.035–0.275	11/13; 0.009
AOM-PLS-DA-global-nipals-adjoint vs PLS-DA	13	0.030	0.000–0.111	8/13; 0.105
POP-PLS-DA-nipals-adjoint vs PLS-DA	13	-0.037	-0.098–0.048	5/13; 0.682
AOMRidgeCls-global-compact vs PLS-DA	14	0.175	0.145–0.264	13/14; 8.5e-04
AOMRidgeCls-branch_global-compact vs PLS-DA	14	0.169	0.137–0.272	13/14; 9.2e-04
AOMRidgeCls-superblock-compact vs PLS-DA	14	0.165	0.124–0.243	14/14; 5.5e-04
AOMRidgeCls-active-compact vs PLS-DA	14	0.163	0.118–0.228	14/14; 5.5e-04
AOMRidgeCls-mkl-compact vs PLS-DA	14	0.163	0.129–0.234	13/14; 8.5e-04

The SIMPLS AOM-PLS-DA variant is the classification row promoted in the main manuscript. NIPALS-adjoint classifiers are less stable. AOM-Ridge classification variants are favourable on the 14-dataset overlap, but they are kept here because the main classification question is the PLS-DA covariance analogue.

12 Seed Stability

AOM-PLS dataset-level RMSEP rankings are highly correlated across seeds, but some datasets change the winning member of the AOM-PLS family. The top AOM Ridge rows are deterministic

Table 13: Seed-stability diagnostics for the multi-seed AOM-PLS and AOM-Ridge runs.

Family	Variant	Full seeds	Mean ρ	Seed CV	Winner changes
AOM-PLS	PLS-standard	32	1.000	0.000	14
AOM-PLS	AOM-compact-cv5	32	0.994	0.017	14
AOM-PLS	ASLS-AOM-compact-cv5	32	0.994	0.021	14
AOM-PLS	AOM-default	32	1.000	0.000	14
AOM-Ridge top5	Ridge-raw	23	1.000	0.000	0
AOM-Ridge top5	AOMRidge global compact	23	1.000	0.000	0
AOM-Ridge top5	AOMRidge local knn50	23	1.000	0.000	0
AOM-Ridge top5	AOMRidge local blended	23	1.000	0.000	0
AOM-Ridge top5	AOMRidge multibranch	23	1.000	0.000	0

for the recorded split/random-state representation. FastAOM was run on seed0 for the wide exploration and should be read as chain-space evidence rather than split-resampling evidence.

Empirical determinism of the AOM-Ridge headline. The main-text AOM-Ridge headline (`AOMRidge-Blender-headline-spxy3` and `AOMRidge-AutoSelect-headline-spxy3`) is reported on the deterministic random-state-0 run, while AOM-PLS, PLS-default, Ridge-default, PLS-HPO and Ridge-HPO use seeds 0/1/2. Two archived audit workspaces re-ran Blender and AutoSelect under seeds 0, 1 and 2 on a partial cohort (`da001_audit20_seeds012` and `da001_partial_fast12_seeds012`); deduplicated by (dataset, variant, seed), their union covers 26 datasets, of which 18 are in the strict main-text denominator $N_\cap = 32$. The two workspaces overlap on 6 datasets; on the 36 cross-workspace (dataset, variant, seed) triples present in both files the maximum absolute RMSEP difference is 0 (assertion enforced in `aggregate_ridge_seed_audit.py`, tolerance 10^{-8}), so the dedup is on values that agree byte-for-byte rather than within numerical tolerance. On every one of the 18 overlap datasets the recorded RMSEP is likewise byte-identical across seeds 0, 1 and 2 for both Blender and AutoSelect (Table 14). The remaining 14 datasets of $N_\cap = 32$ were not included in the archived audit workspaces and are outside the scope of this check; whether their RMSEP would also be seed-invariant cannot be asserted from the present evidence. This identity is largely a consequence of the protocol: the `SPXY3` split used for the headline only consumes `random_state` when a PCA pre-projection is requested, and the Blender/AutoSelect inner selectors then condition on the deterministic split. The observation should therefore be read as *protocol determinism*, not as evidence of robustness to repeated train/test partitions. The main text uses "comparable to" or "improves over" rather than "robust across seeds" when discussing the AOM-Ridge headline ratios for this reason. A repeated partition sensitivity check on a stratified subset of datasets is listed as an open extension in the main-text discussion.

Table 14: AOM-Ridge headline seed audit on the 18-dataset overlap between $N_\cap = 32$ and the archived `da001_audit20_seeds012/da001_partial_fast12_seeds012` workspaces. "Max RMSEP span across seeds" is the per-dataset maximum $\max_s \mathrm{RMSEP}(s) - \min_s \mathrm{RMSEP}(s)$, aggregated as the row maximum. Both selectors record identical RMSEP at seeds 0, 1 and 2 on every dataset.

Variant	Audit datasets	Seeds 0/1/2 complete	Max RMSEP span across seeds	Datasets with non-zero span
AOMRidge-AutoSelect	18	18	0	0
AOMRidge-Blender	18	18	0	0

13 Reproducibility and Software Artifacts

Table 15: Software artifacts distributed with the AOM study.

Component	Tests / evidence	Status
`nirs4all-aom` Python package	AOM-PLS, POP-PLS, AOM-Ridge and FastAOM implementations, with unit tests and sklearn-compatible APIs.	public software
`nirs4all-aom` benchmark artifacts	Benchmark runners, result CSVs, aggregation scripts, figure builders and manuscript tables used here.	release with paper
`nirs4all` instrumentation context	NIRS instrumentation, acquisition and provenance context for local benchmark inputs.	instrumentation software
Supplementary validation dossier	Cohort manifest, missing-dataset audit, paired statistics and software-readiness notes distributed with `nirs4all-aom`.	public evidence

All numerical claims in the main manuscript and in this supplement are computed from benchmark result tables distributed with the public `nirs4all-aom` repository announced in the main manuscript. The regression cohort contains 61 manifest rows, and the main paired regression comparisons use $N_\cap = 32$ rows. The classification manifest contains 17 rows, with 16 included in benchmark analyses after one missing-file exclusion, and is summarized in the classification tables.

Multi-seed evidence is aggregated at dataset level before paired tests. AOM-PLS, PLS-default, Ridge-default, PLS-HPO and Ridge-HPO use seeds 0, 1 and 2. The AOM-Ridge headline selectors are reported on the deterministic headline run with random state 0, while the supplementary seed-stability table reports the multi-seed AOM-PLS and AOM-Ridge diagnostics. The main regression comparisons use paired dataset-level RMSEP ratios, Wilcoxon signed-rank tests where applicable and Holm correction within the reported comparison family. Classification comparisons use paired balanced-accuracy differences with the same family-wise correction. The validation status of the software artifacts is summarized in Table 15.

14 Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Anthropic Claude and OpenAI Codex to assist with code review, implementation, repository organization, benchmark aggregation scripts, LaTeX editing and drafting or revision of explanatory text. Every numerical claim in the main manuscript and in this supplement is computed from benchmark result tables; the AI-assisted tools did not generate numerical results by themselves. After using these tools, the authors reviewed, edited and verified the code, numerical results, references and manuscript content as needed, and take full responsibility for the content of the publication.

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